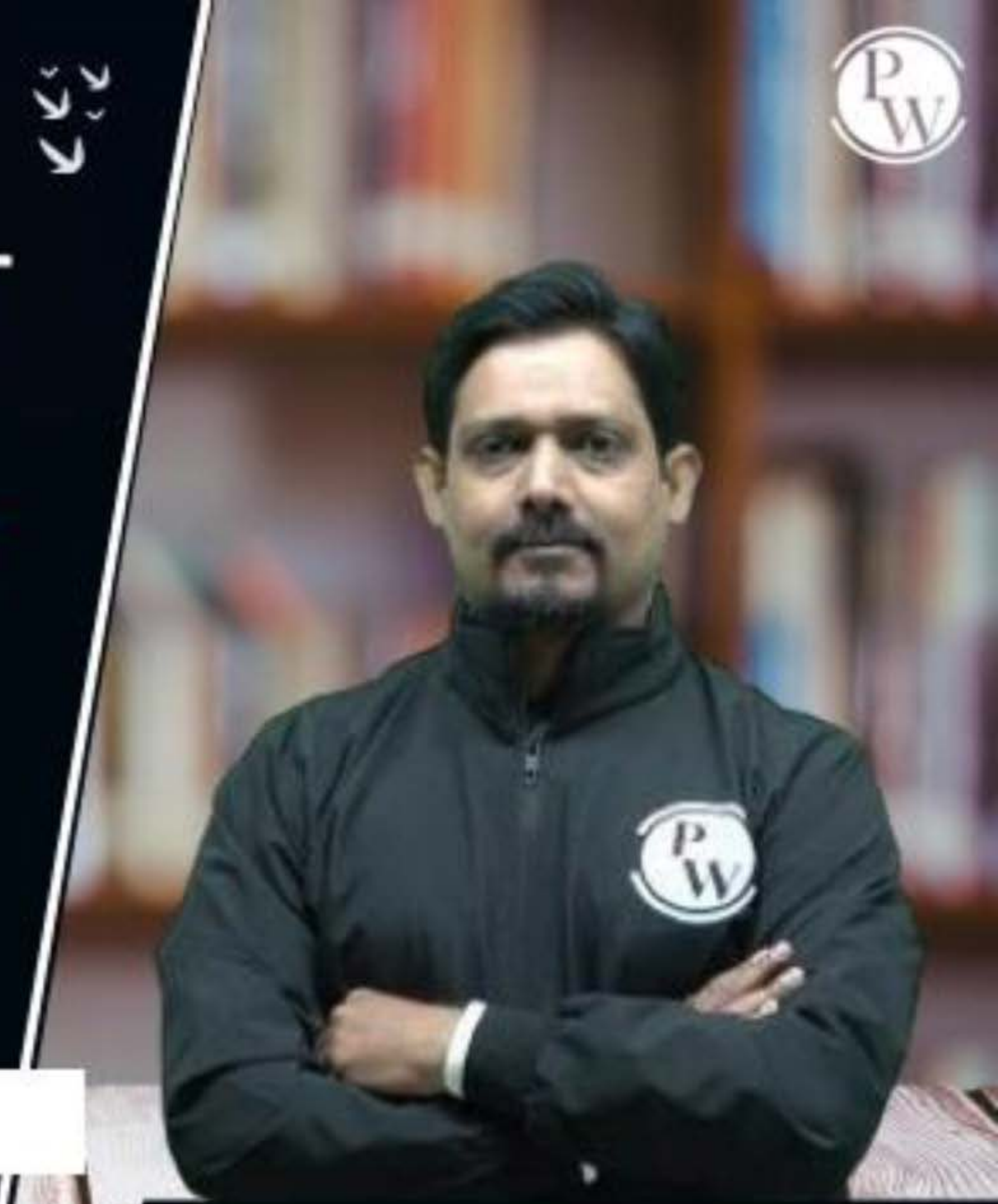
Computer Science & IT

Data Structure & Programming

Tree

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture

- Topic — Counting of labeled tree
- Topic — Representation of tree
- Topic — Traversal
- Topic — Construction of tree using In order & preorder
- Topic

Slide

Topics to be Covered

Topic Tree construction using postorder - Inorder

Topic Tree construction using post and pre (special case)

Topic BST

Topic BST Insertion

Topic BST traversal

Slide

Building tree from post *and in- order*

Construction of binary tree from preorder and inorder traversal.

Postorder: *c f g d b e a*

Inorder: *c b f d g | a | e*

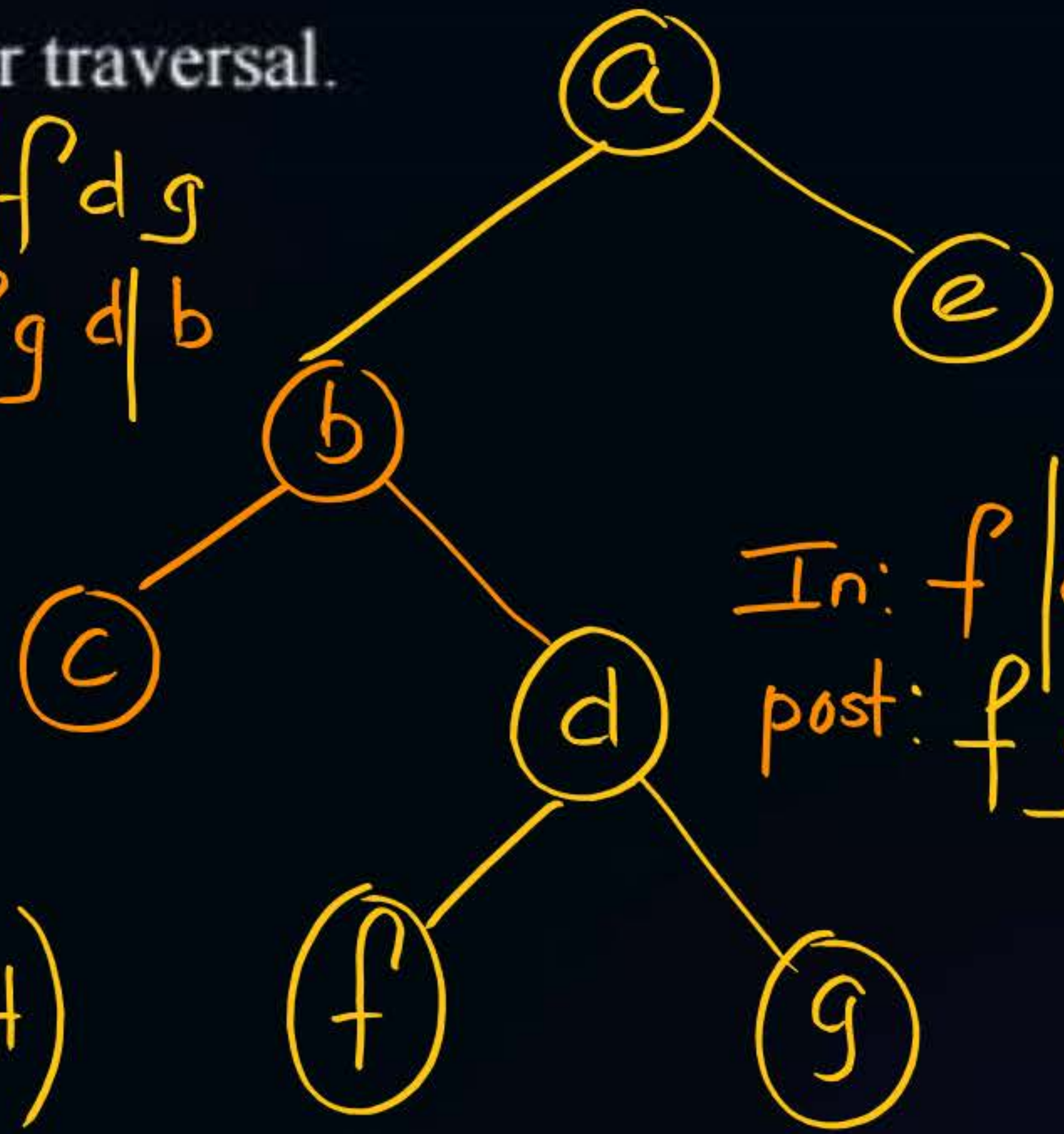
In : c | b | f d g
post : c | f g d | b

Left Right Root

Left Root Right

(I) Identify the Root
in post order (Last element)

(II) use Root to divide
Inorder in Left and right subtree



In: f | d | g
post: f g d

Topic : Question

#Q if the inorder traversal is 2, 7, 5, 6, 11, 1, 9, 5, 8

Post order : 2, 5, 11, 6, 7, 5, 8 9, 1. If binary tree constructed then sum of leaf nodes is

[23]

In: 2,7,5,6,11

post: 2,5 11,6,7

In: 9,5,8

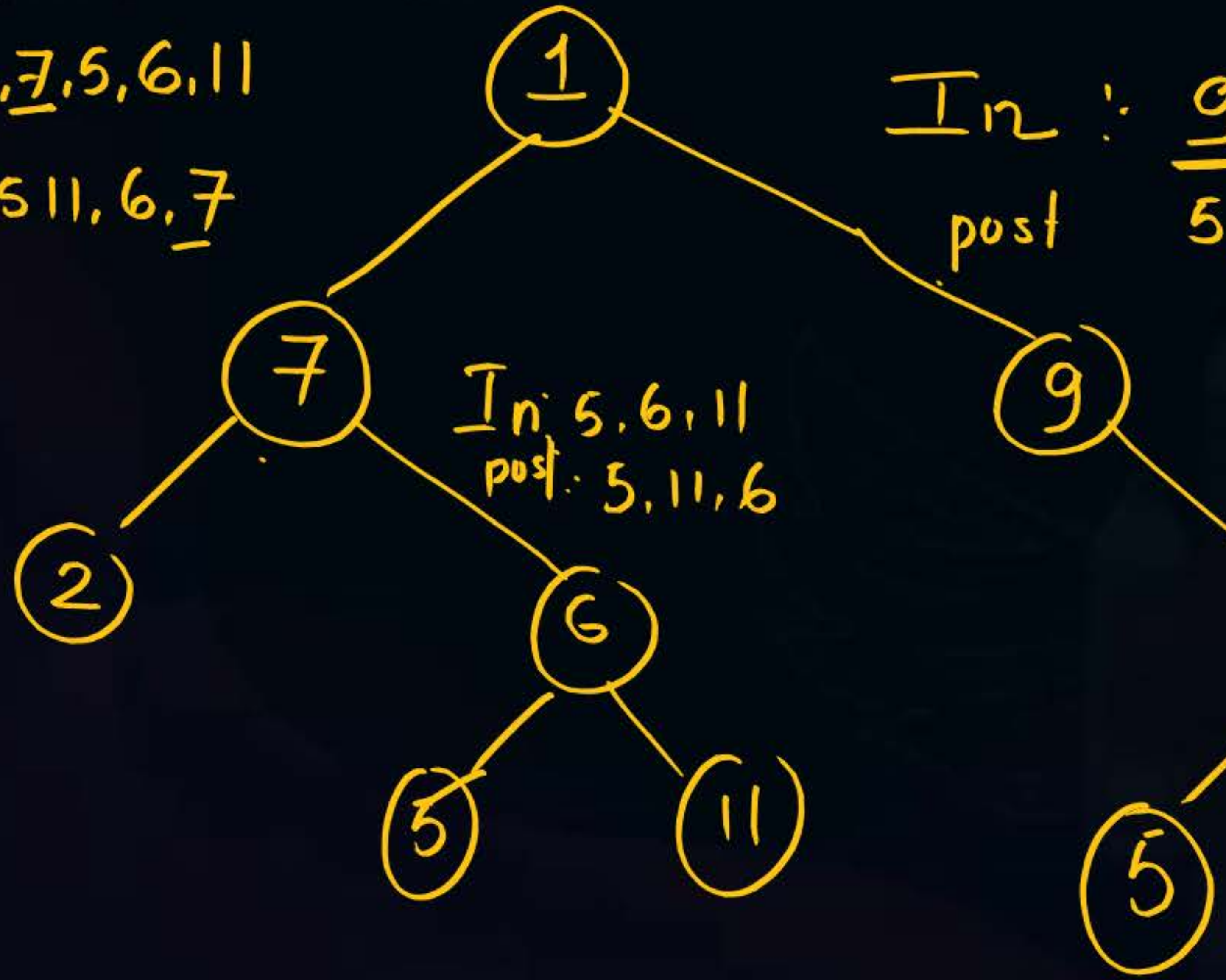
post 5,8,9

In: 5,6,11

post: 5,11,6

In: 5 8

post: 5,8



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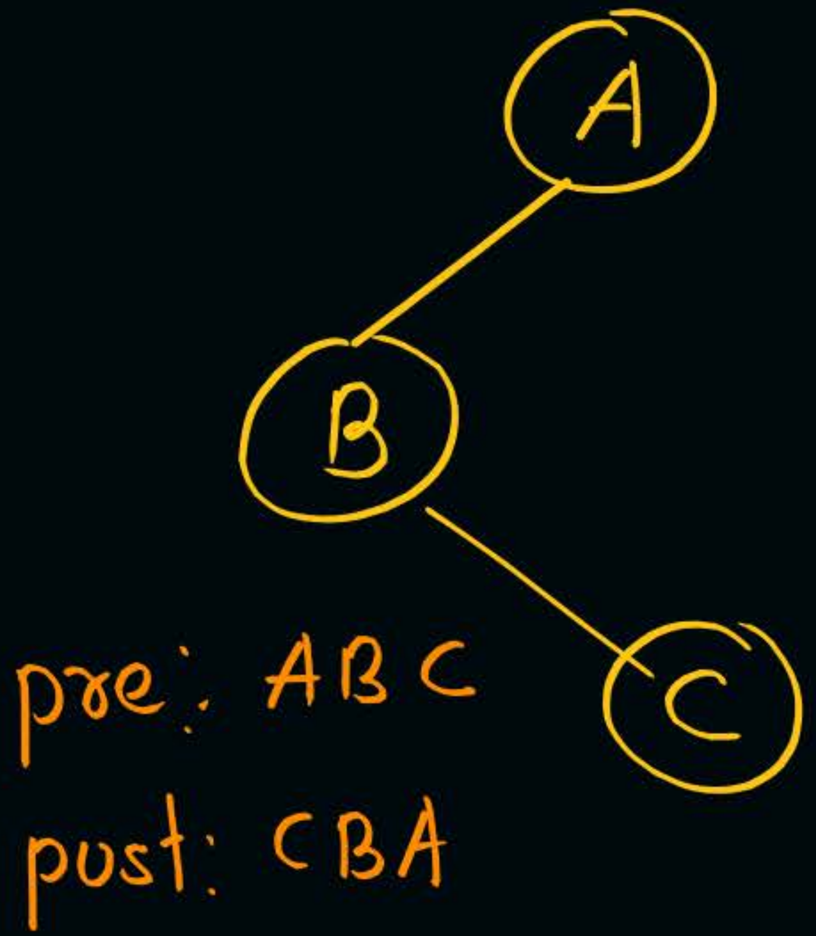
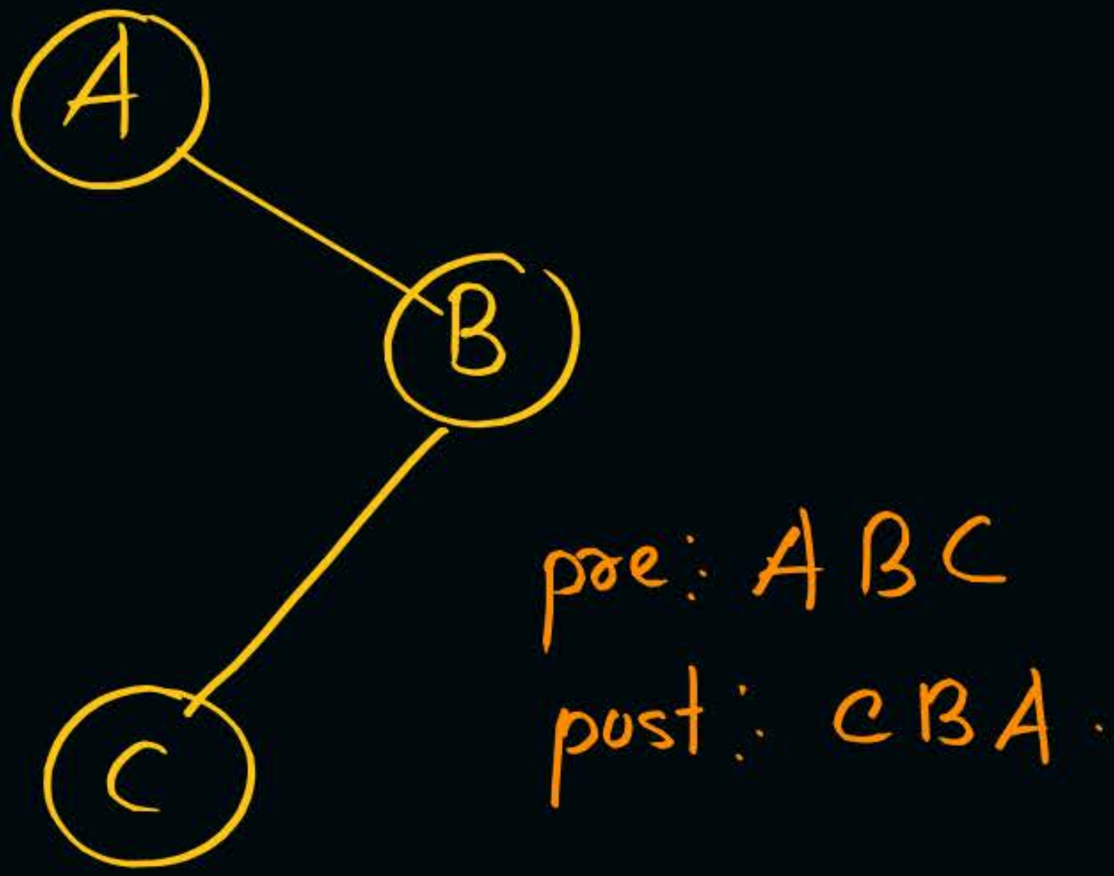
post and pre order





Unique tree not possible

Topic : Binary Search Tree

Binary search tree is a tree in which $\forall$ node with key value lesser than parent Node goes to Left subtree and all nodes with key value greter than parent node goes to right subtree.

This property subsequently followed in subtree as well

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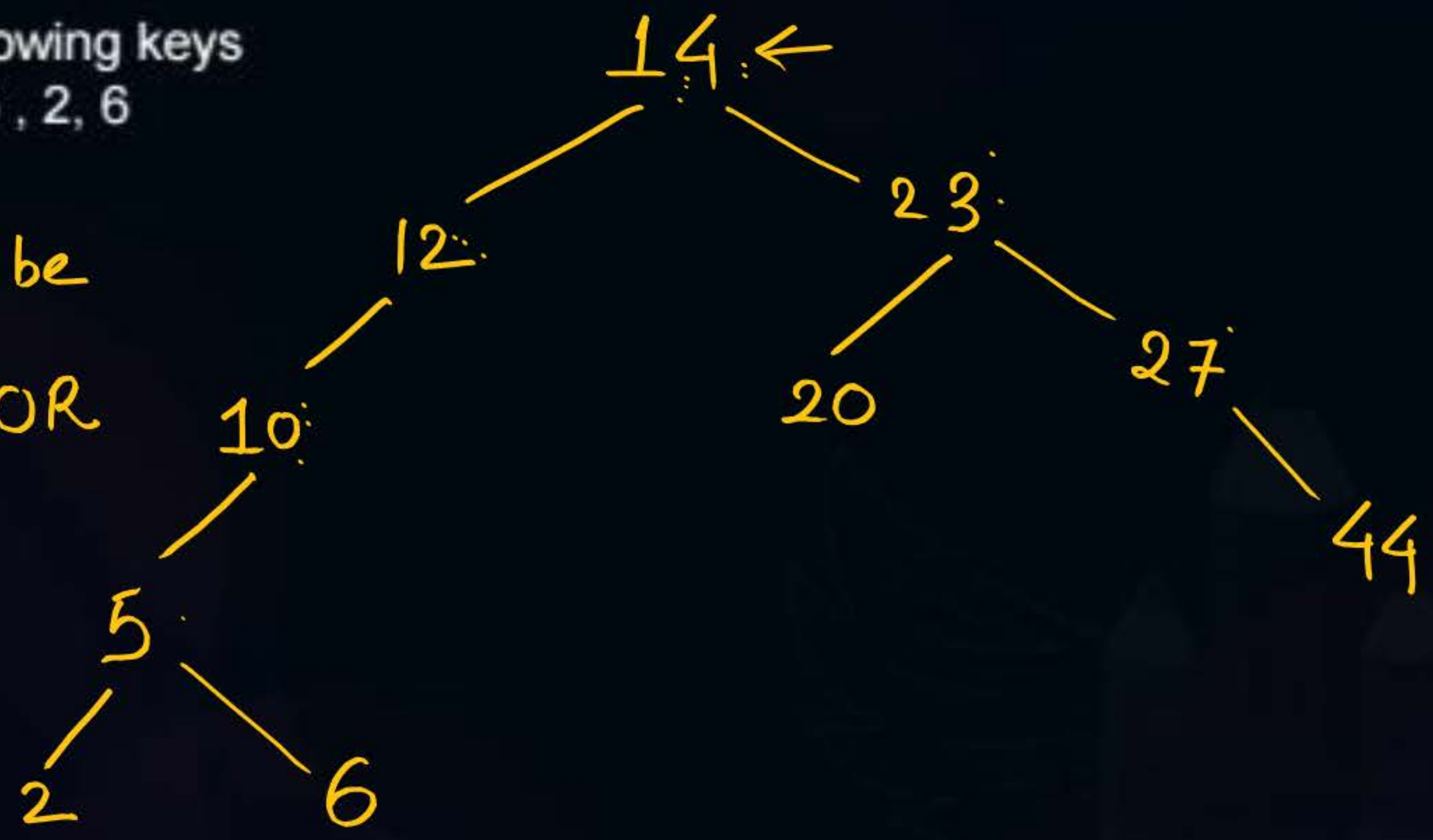
Topic : Construction of Binary Search Tree

Suppose We insert the following keys
14, 12, 23 ,10 27, 20, 44, 5 , 2, 6

equal key can be present in left OR right subtree



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Topic : Construction of Binary Search Tree

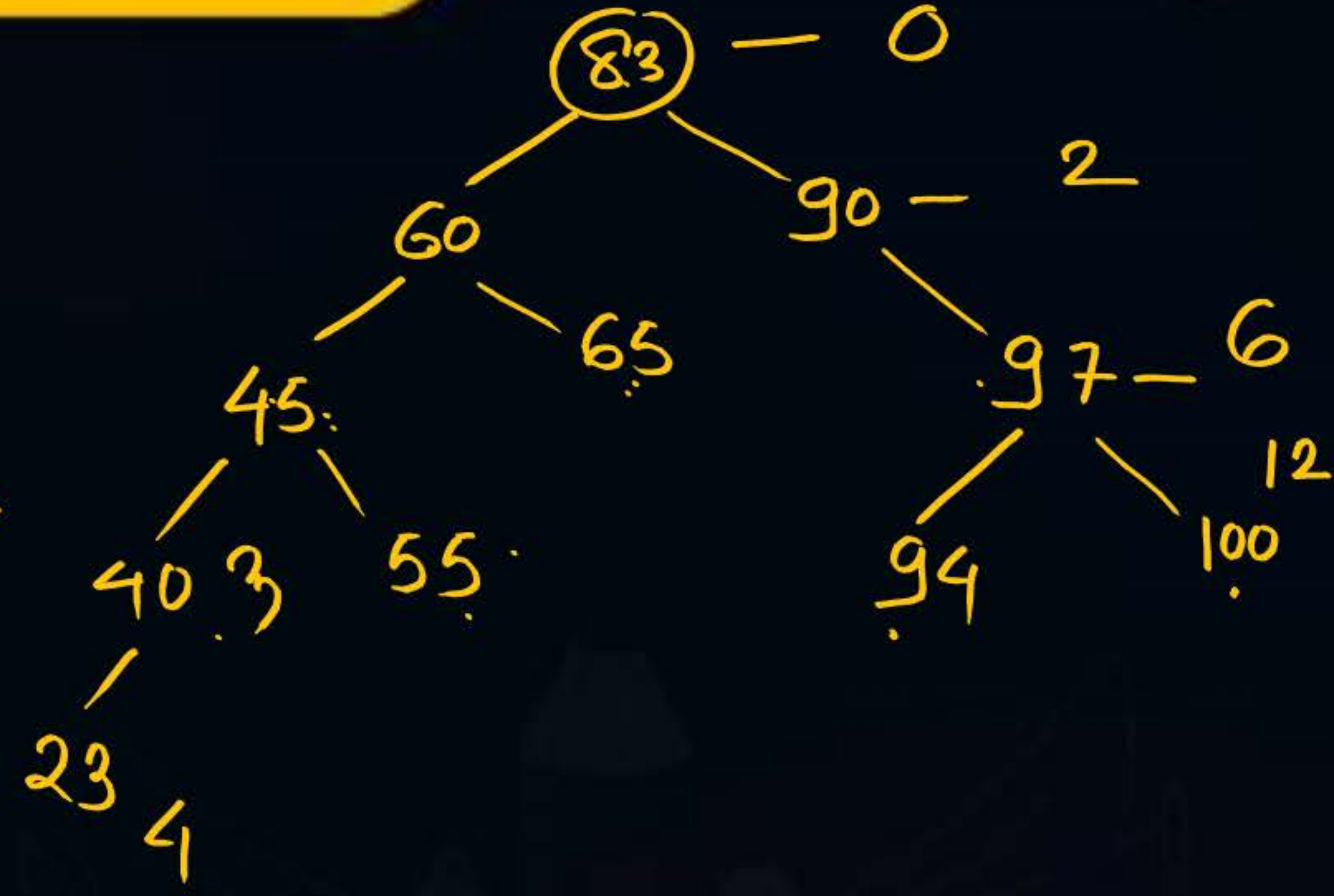
Suppose We insert the following keys

83, 60, 90, 45, 97, 100, 94, 55, 65, 40, 23

The number of comparison required for construction of it

first element : Zero comparison

24 comparison



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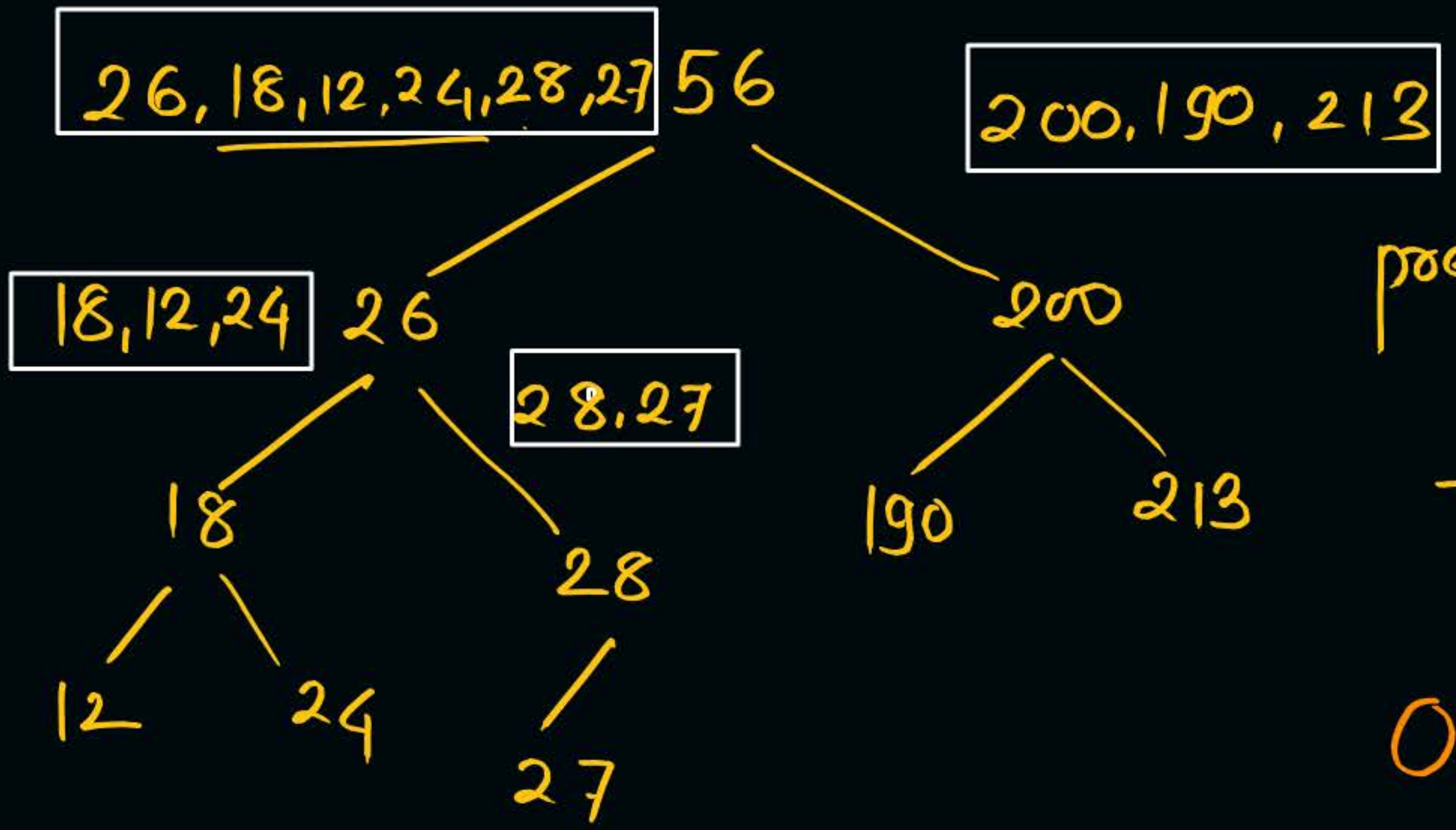
Inorder : 12, 18, 24, 26, 27, 28, 56, 190, 200, 213

Root can't Identified

∵ Inorder is Increasing order sequence

BST from preorder traversal



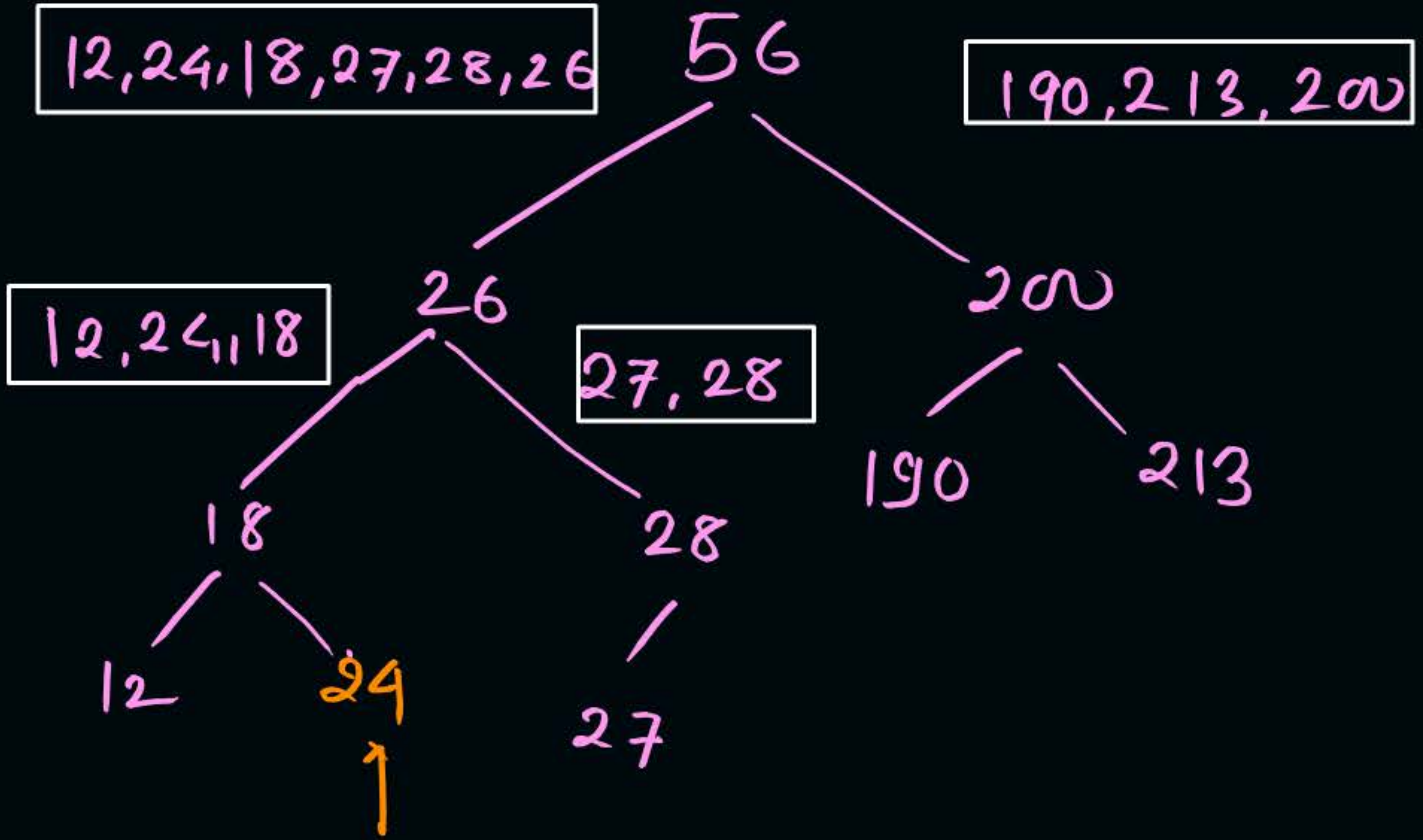
preorder: 56 $\boxed{26, 18, 12, 24, 28, 27}$, $\boxed{200, 190, 213}$

Lesser

first Node is Root, Left and Right Subtree

Only preorder BST can be constructed

BST from post order



postorder: 12, 24, 18, 27, 28, 26, 190, 213, 200, 56

Left subtree

Right subtree

Topic : Question

2-ary tree (0 - 2 children)

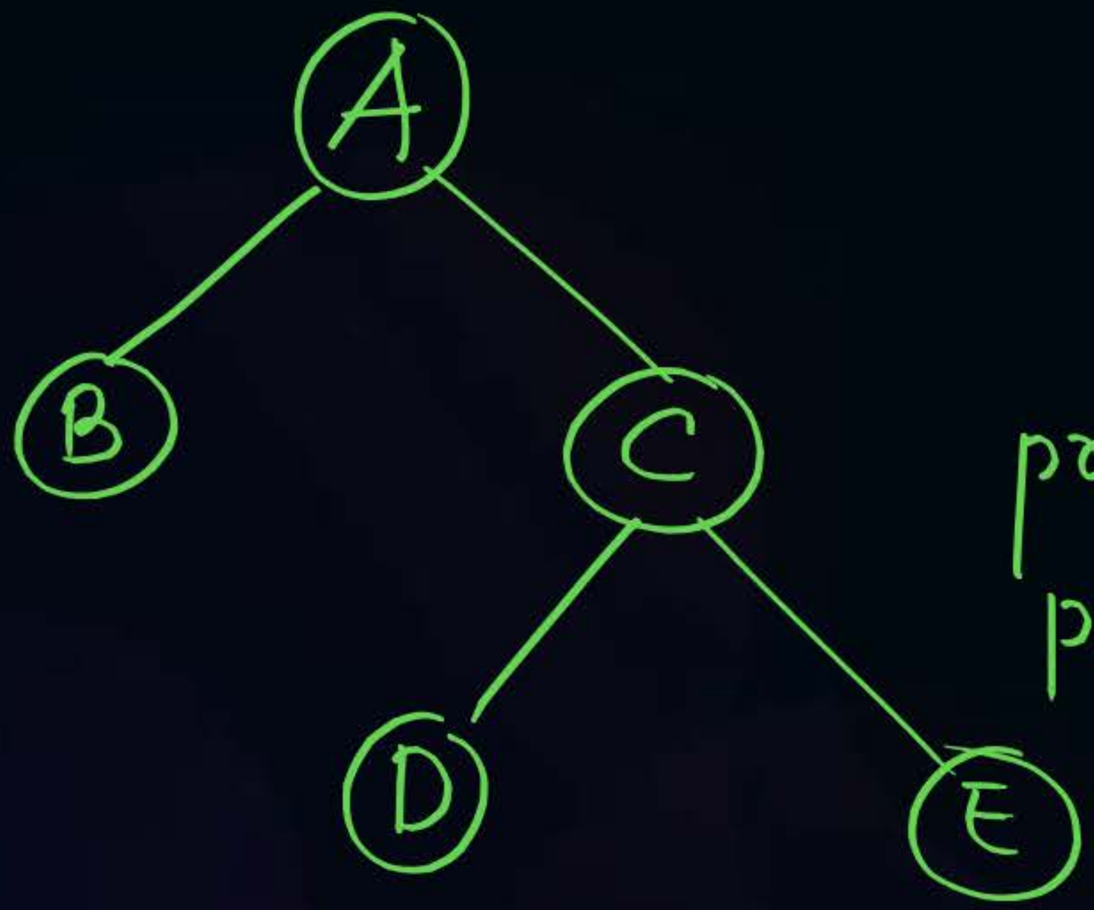
Root of Left subtree

pre : A | B | C D E   Root Left Right

post : B | D E C | A

Left Right Root

Root of Right subtree



pre : C D E

post D E C

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Tree with each node 0 or 2 children.

Left subtree Root

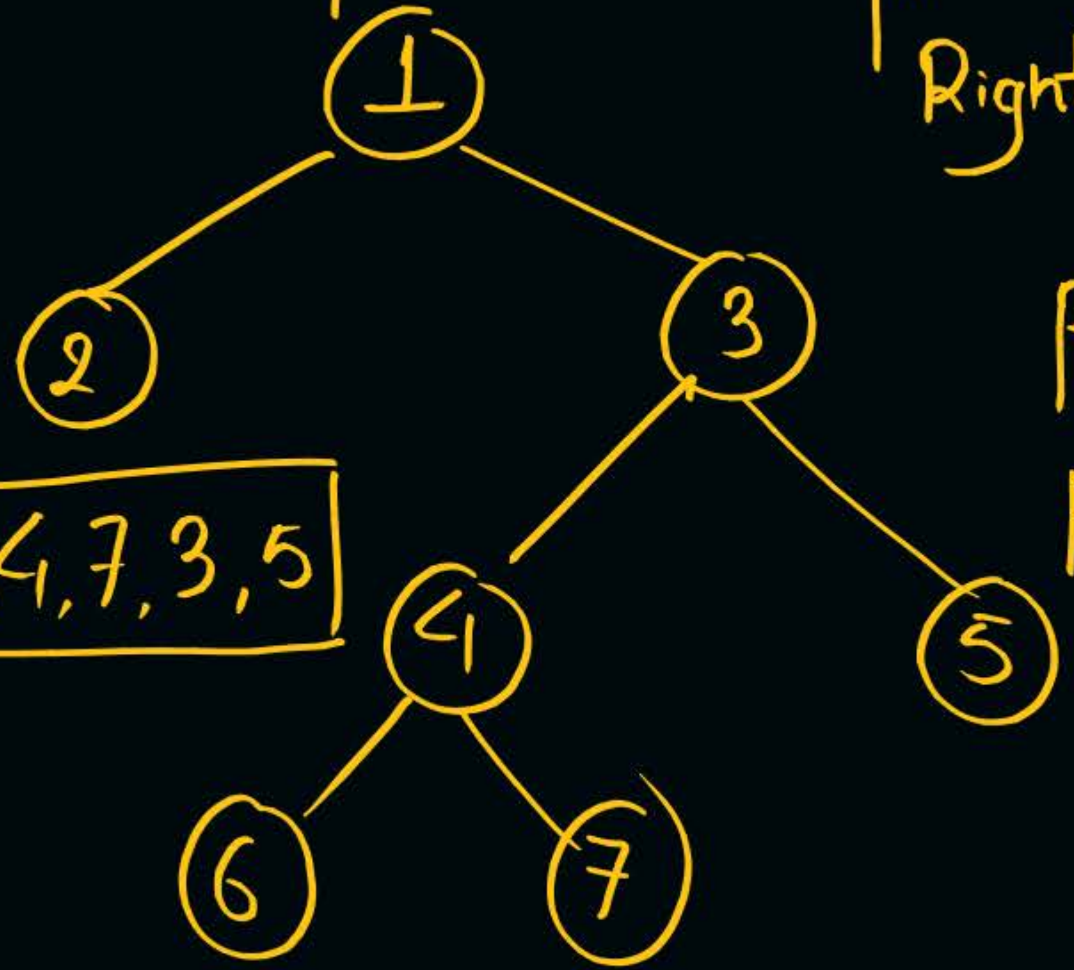
pre : 1, | 2, | 3, 4, 6, 7, 5

post : 2, | 6, 7, 4, 5, 3 | 1

Right subtree Root

Root <u>left</u> Right

left right Root



$\boxed{2\ 1, 6, 4, 7, 3, 5}$

pre : 3, | 4, 6 7, | 5

post : 6, 7, 4 | , 5 | 3

Topic : Question

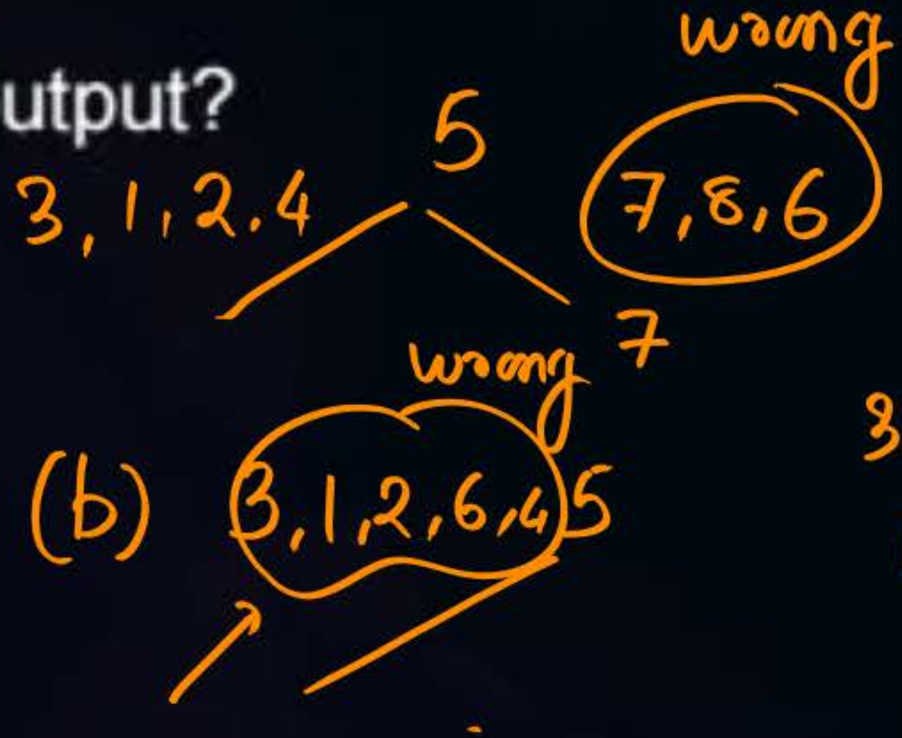
A binary search tree contains the value 1, 2, 3, 4, 5, 6, 7, 8. The tree is traversed in pre-order and the value are printed out. Which of the following sequence is a valid output?

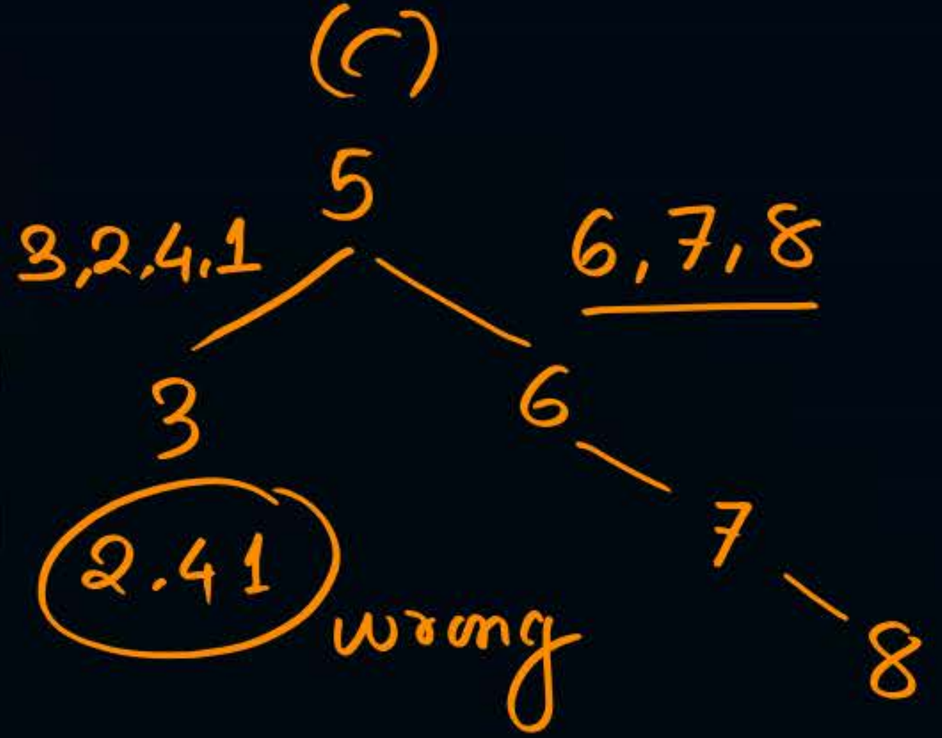
(a) 5 3 1 2 4 7 8 6

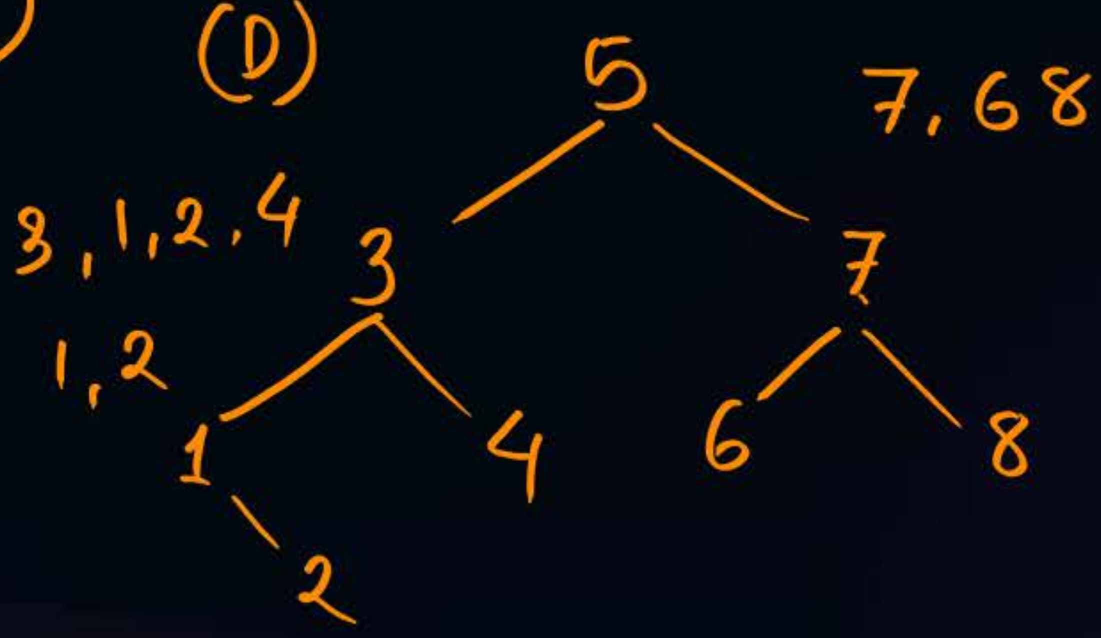
(b) 5 3 1 2 6 4 8 7

(c) 5 3 2 4 1 6 7 8

(d) 5 3 1 2 4 7 6 8







Topic : Binary Search Tree

MSQ

#Q The binary search tree shown below was constructed by inserting a sequence of items into an empty tree

Which of the following input sequences will produce this binary search tree?

(a) 5 3 4 9 12 7 8 6 20 ✓ (A)

(b) 5 93 768 4 12 20 ✓

(c) 5 9 738 12 6 420 ✓

(d) 5 93678 4 1220

(b)

(c)

Slide

Topic : Question

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#Q. You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T. and you do not have access to T.

Which one of the following statements is TRUE?

(A) Postorder traversal of T can be determined from V ✗

(B) Inorder traversal of T can be determined from V ✓ Increasing order sequence

(C) Root node of T can be determined from V ✗

(D) Preorder traversal of T can be determined from V ✗

Slide

Topic : Question

#Q. For given preorder : ABCDEFG, InOrder: BDCAFGE

What is the postorder of Binary Tree

(A) CDBGFEA

(B) DBCGFEA

(C) DCBGFEA ✓

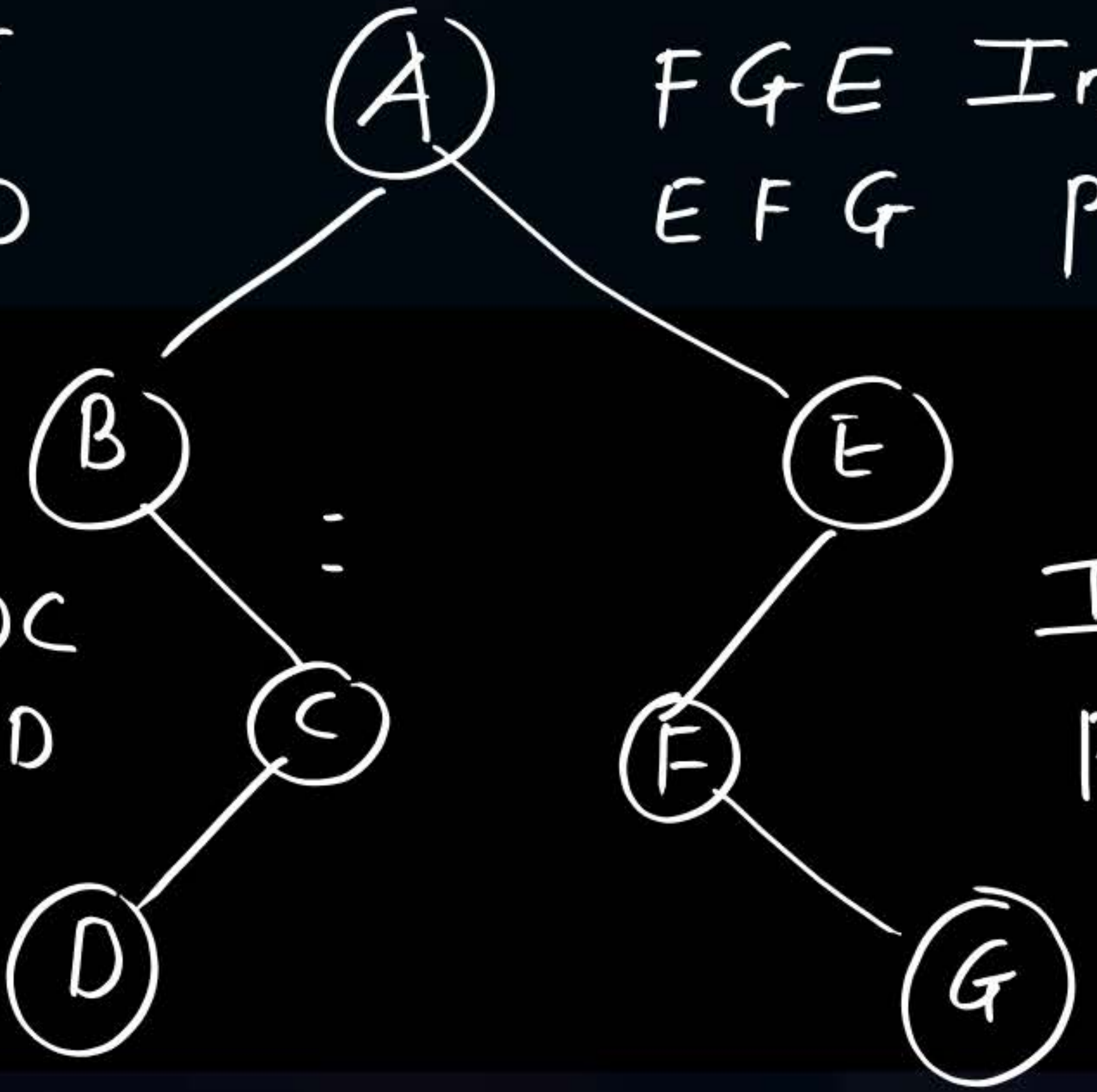
(D) None of these

In BDC

pre BCD

FGE In

EFG pre



In: DC

pre CD

In: FG

pre F F

DCBGFEA

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THANK - YOU

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